Receiver-selective visual writing | LLaVA-1.5-7B / GQA3000

A Which tokens receive vision?

On which side is the man?

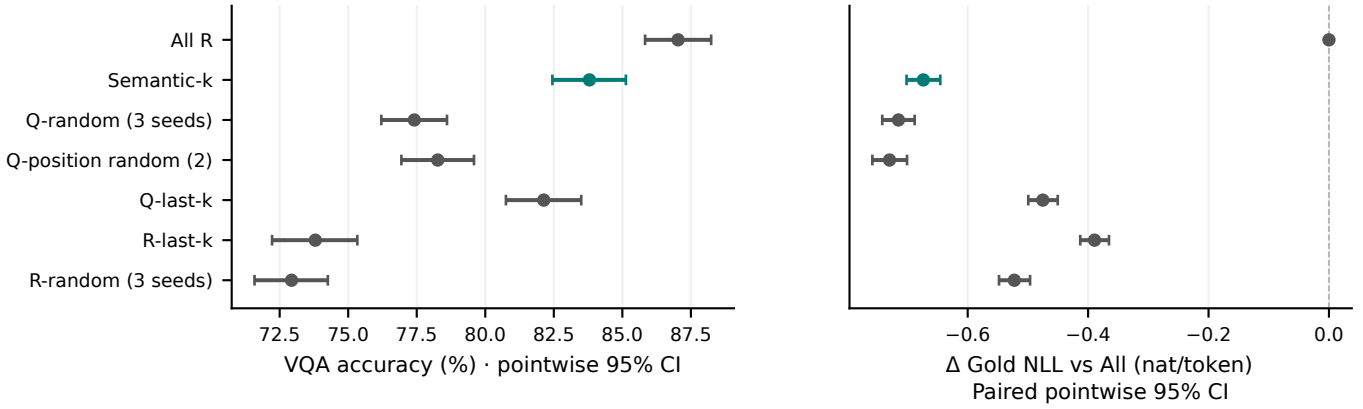
Fixed example: GQA 00150746

Semantic	On	which	side	is	the	man	?
Question-last	On	which	side	is	the	man	?
Question random	On	which	side	is	the	man	?

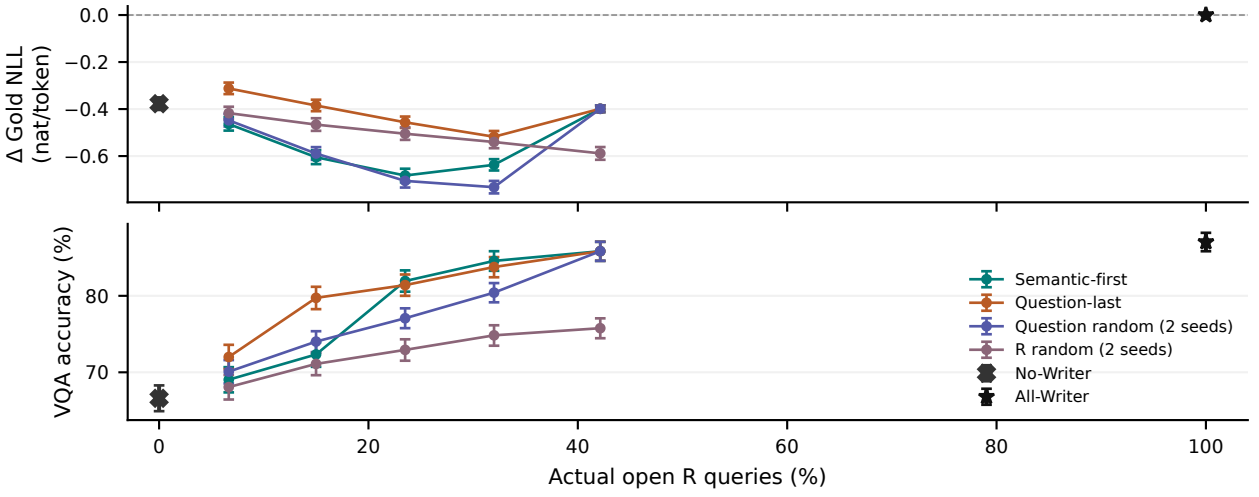
Each row: 4 tokenizer tokens / 24 R queries. Whole words stay intact. R also includes instruction/template tokens.

B Receiver selection matters

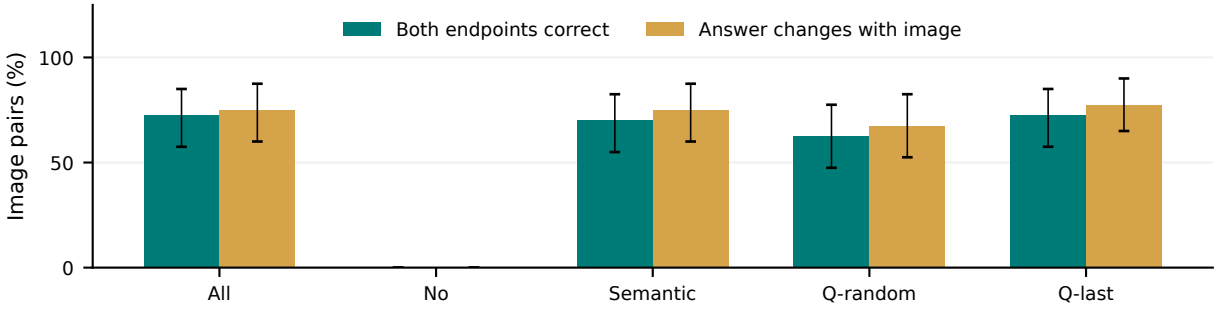
Semantic - Q-random: ΔAcc +6.39pp [+5.35, +7.41]
ΔNLL +0.041 [+0.023, +0.059] (paired 95% CI)



C Token budget-quality frontier



D Does sparse writing preserve grounding?



40 frozen different-Gold image pairs; linked-pair 95% CI.
No-writer: all effective logits strictly identical across images.

All answer queries block direct image access at all 32 layers. Curve budgets are matched by whole-word token counts.
Curve intervals are pointwise. Confirmatory tests use a separate prespecified Bonferroni family.